

Unit 2 Pre-Test

Exponent Rules & Properties

Standard assessed: NC.8.EE.1

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.EE.1 Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

• *I can apply the properties of integer exponents to simplify expressions with numerical bases.*

Match each vocabulary term to its definition, then simplify each expression.

D ▾ base ✓	A. The numerical factor multiplied by the variable(s) in a term.
B ▾ exponent ✓	B. Tells how many times the base is used as a factor.
A ▾ coefficient ✓	C. Terms with identical variable parts (same variables to the same powers).
F ▾ monomial ✓	D. A number or variable used as a repeated factor in a power.
C ▾ like terms ✓	E. An expression that uses a root symbol, such as a square root ($\sqrt{}$).
E ▾ radical ✓	F. A single term: a number, a variable, or a product of numbers and variables.

Parts of a monomial — reference

$$5x^3$$

coefficient the number factor	base the repeated factor	exponent how many times the base is a factor
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A **monomial** is a single term — a number, a variable, or a product such as the one above.

1. Simplify:

$$4^3 \cdot 4^2$$

Answer:

answer: 4^5

2. Simplify:

$$(5^2)^3$$

Answer:

answer: 5^6

3. Simplify:

$$\frac{6^7}{6^4}$$

Answer:

answer: 6^3

4. Simplify:

$$2^4 \cdot 2^4$$

Answer:

answer: 2^8

5. Simplify:

$$(10^3)^2$$

Answer:

answer: 10^6

6. Simplify:

$$\frac{9^5}{9^2}$$

Answer:

answer: 9^3

SCORE 4.0 Target Content

NC.8.EE.1 Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

• *I can multiply and divide monomials using the properties of exponents.*

Multiply, divide, or raise to a power. Simplify each monomial expression.

7. Multiply:

$$2x^3 \cdot 5x^4$$

Answer:

answer: $10x^7$

8. Multiply:

$$(4a^2b)(3a^3b^5)$$

Answer:

answer: $12a^5b^6$

9. Divide:

10. Divide:

$$\frac{12x^6}{3x^2}$$

Answer:

answer: $4x^4$

$$\frac{20a^7b^4}{5a^3b^2}$$

Answer:

answer: $4a^4b^2$

11. Simplify:

$$(2x^3)^4$$

Answer:

answer: $16x^{12}$

12. Simplify:

$$(3x^2y^4)^3$$

Answer:

answer: $27x^6y^{12}$

SCORE 5.0 Advanced Content

NC.8.EE.1 Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

• *I can simplify complex, multi-step monomial expressions, including zero and negative exponents.*

Simplify using several exponent properties. Write answers with positive exponents.

13. Simplify:

$$\frac{(2x^3y^4)(5x^2y)}{10x^5y^5}$$

Answer:

answer: 1

14. Simplify:

$$\frac{4a^5b^3}{4a^5b^7}$$

Answer:

answer: $\frac{1}{b^4}$

15. Simplify:

$$(2x^{-2})^3$$

Answer:

answer: $\frac{8}{x^6}$

16. Simplify:

$$(4x^3y^{-2})(2x^{-5}y^4)$$

Answer:

answer: $\frac{8y^2}{x^2}$

17. Simplify:

$$\frac{(3x^2y^4)^2}{9x^5y^6}$$

18. Simplify:

$$\frac{(6x^5y^2)(2xy^3)}{3x^4y^7}$$

Answer:

answer: $\frac{y^2}{x}$

Answer:

answer: $\frac{4x^2}{y^2}$